Lecture 29

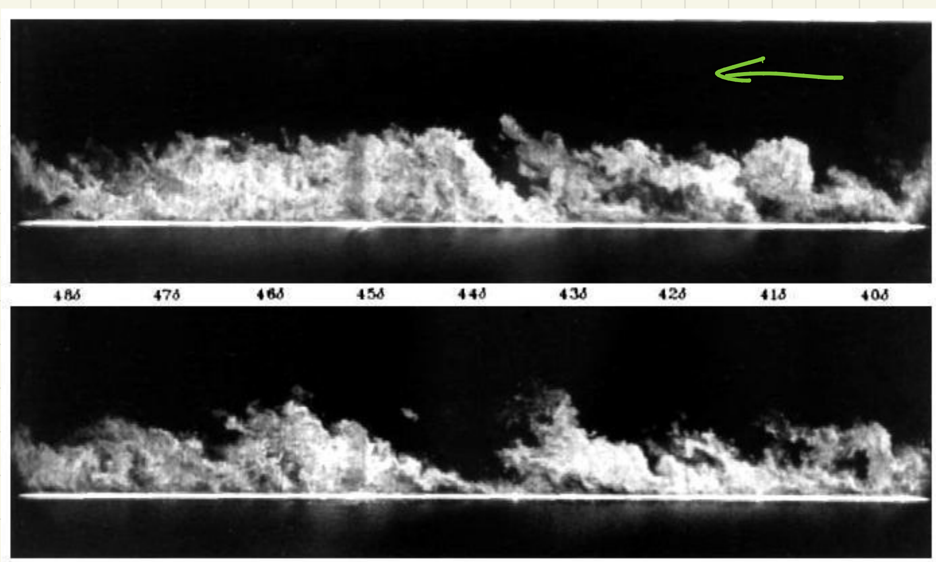
Turbulence

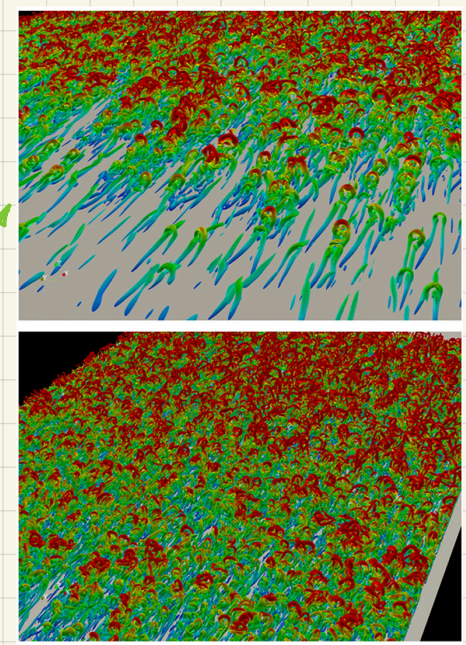
Turbulence (in boundary layers)

PLAN
- general overview
- BL – generically intractable – some amazing results

Turbulence

- a solution of N-S equations
  - so physics is known in gov. eq. sense
  - hard to describe / reduce – problem of description or prediction





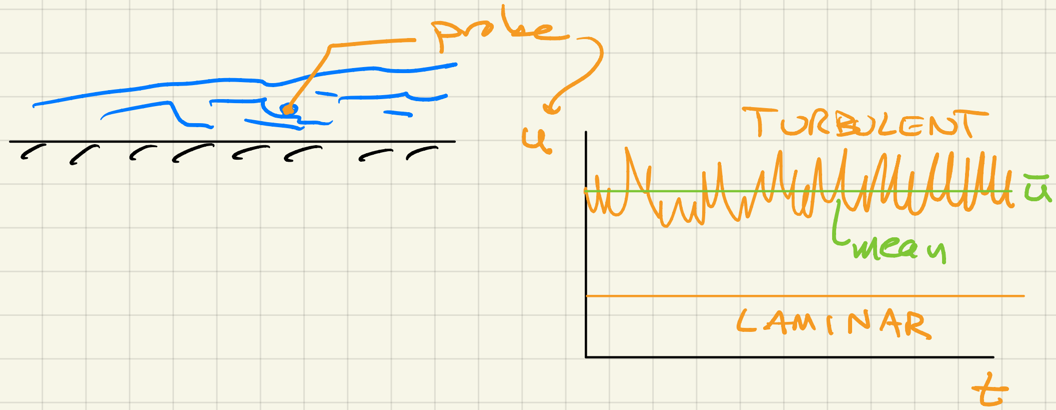
λ_2 criterion

• unsteady, random, seemingly chaotic, 3D → fluctuations

true turbulence is 3D → vortex stretching term in vorticity equation

$\underset{\sim}{u} \cdot \nabla \underset{\sim}{\omega}$

$= 0$ in 2D



• nonlinearly active fluctuations

$$\underset{\sim}{u} = \overline{\underset{\sim}{u}} + \underset{\sim}{u}'$$

mean/average

fluctuation

can't neglect $|\underset{\sim}{u}'|^2$ terms....

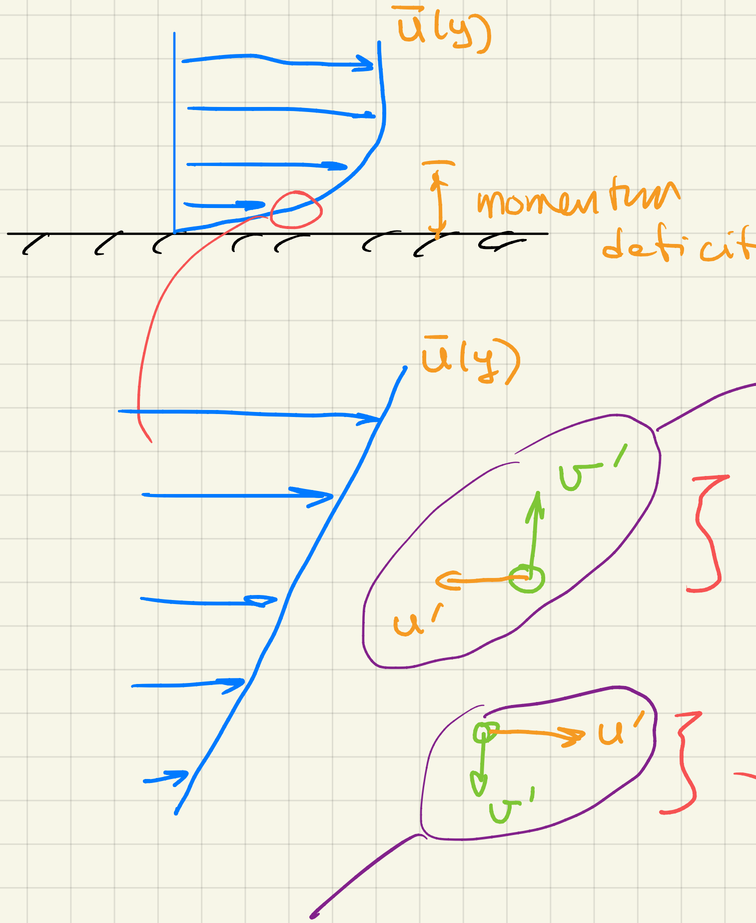
$$\frac{u'_{rms}}{\bar{u}} = 0.1 \text{ in BL}$$

0.2 free shear flow

→ small enough that much of the laminar flow phenomenology still applies

→ but $\overline{u'u'}$ terms affect details enough to change the flow quantitatively

- fluctuations transport: momentum, heat, species, ...



Laminar case → viscosity transports momentum (and vorticity)

+y transport of -x momentum

if $v' > 0$, that bit of fluid is coming from a smaller $\bar{u}$ region: expect $u' < 0$

if $v' < 0$, that bit of fluid is coming from a higher $\bar{u}$ region: expect $u' > 0$

-y transport of +x momentum

like the flow streaming

- streaming was deterministic
- statistical

⇒ analogous to molecular transport

$\Rightarrow \overline{u'v'} < 0$ will "show up" as an effective stress on $\overline{u}$

- turbulence is associated with instabilities
  - instability: small perturbations can amplify to lead to turbulence – linear mechanism
  - some linearly stable flows are turbulent

- turbulence is self sustaining
  - once turbulent, can keep extracting turbulence fluctuation energy (turbulence kinetic energy) from mean flow kinetic energy

viscosity: $\quad KE \longrightarrow$ heat

$\sim u^2$

turbulence $\quad KE \longrightarrow TKE \xrightarrow{\text{viscosity}}$ heat

$\sim \underset{\sim}{\overline{u}}^2 \qquad \sim \overline{u'u'}$

turbulence kinetic energy

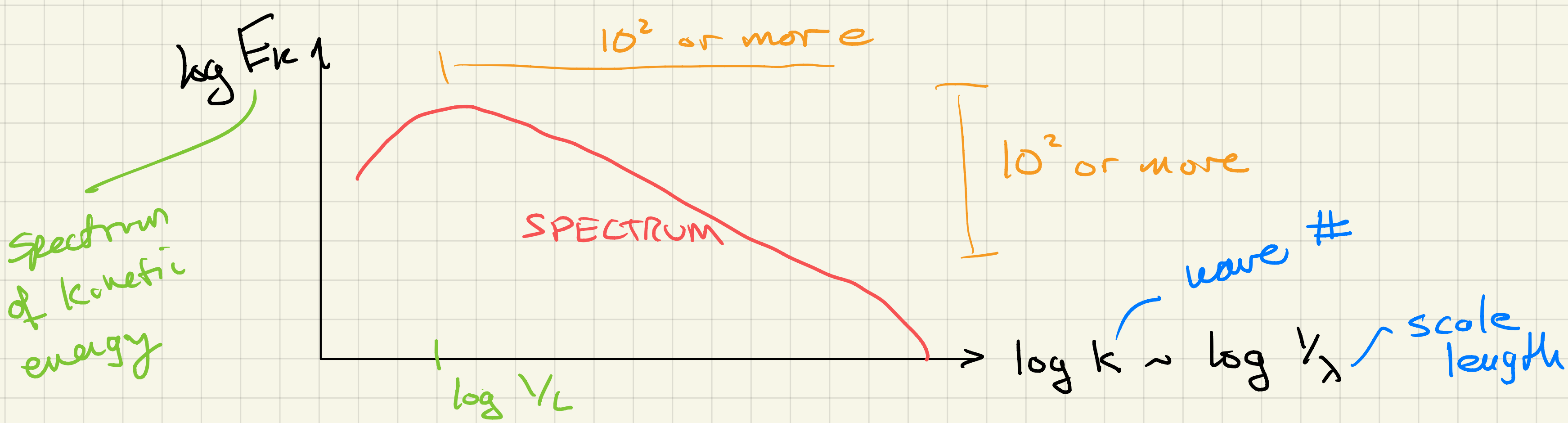
- range of length scales
  - flow scale L $\quad Re_L \gg 1$
  - viscous scale (Kolmogorov scale) η $\quad Re_\eta \sim 1$

loosely like $Re_\delta \approx 1$

larger Re_L is $\rightarrow$ larger the range of scales

L/η increases

- continuous range of scale

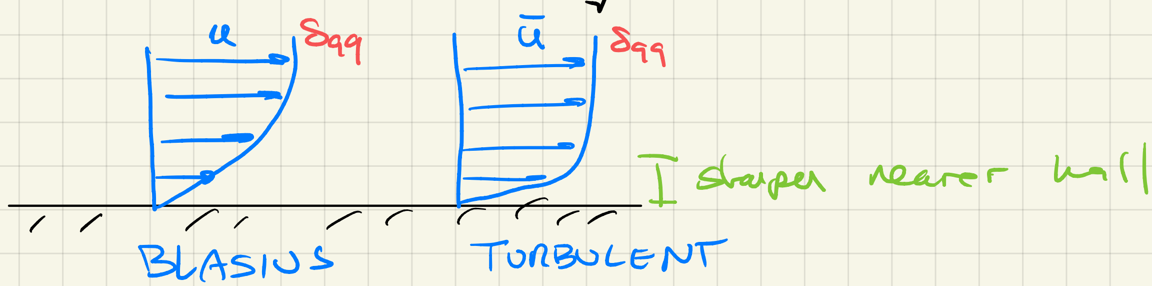


- human scale $1m$ $1m/s$ $\nu = 10^{-5}\ m^2/s$

$Re = 10^5 \rightarrow$ always turbulent

- BL rare exception where laminar flow can persist into engineering Reynolds numbers

- turbulent BL's functionally resemble laminar BL



- generally interested in mean quantities - time averages
  - average drag
  - average heat flux

... but depend on fluctuations

- example: mean + fluctuations

LINER gov. eq. $u = \xi - 1$ $\xrightarrow{\text{average}}$ $\bar{u} = \bar{\xi} - \bar{1}$

ξ random process 0 or 1

$\bar{\xi} = 1/2$

$\bar{u} = -1/2$

NONLINER gov. eqn $u^2 + 2u = \xi - 1$ $\xrightarrow{\text{average}}$ $\overline{u^2} + 2\bar{u} = \bar{\xi} - 1$

like turbulence

desired mean

two unknowns $\overline{u^2} \neq \bar{u}^2$

can solve in this case...

$$u = \frac{1}{2}\left(-2 \pm \sqrt{4 - 4(1-\xi)}\right)$$

$$u = -1 \pm \sqrt{\xi}$$

assume only physical solution e.g. $u > -1$

$$u = -1 + \sqrt{\xi}$$

$$\bar{u} = ?$$

$$\bar{u} = \frac{1}{2}\left(-1+\sqrt{1}\right) + \frac{1}{2}\left[-1+\sqrt{0}\right]$$

half the time

half the time

unclosed

1 eq, 2 unknowns

$$\bar{u} = -1/2$$

$$u^2 = \left(-1+\sqrt{\xi}\right)^2$$

$$\overline{u^2} = \frac{1}{2}\left(-1+\sqrt{1}\right)^2 + \frac{1}{2}\left(-1+\sqrt{0}\right)^2$$

$$\overline{u^2} = \frac{1}{2}$$

assume a model: $\overline{u^2} = a\,\bar{u}^2$

model coefficient

for turbulence we'd need $\overline{uu}$ in terms of $\bar{u}$

$$\bar{u}^2 + 2\bar{u} = \bar{\xi} - 1$$

$$a\,\bar{u}^2 + 2\bar{u} = \bar{\xi} - 1 \quad -1/2$$

only

$$\bar{u} = \frac{-2 \pm \sqrt{4 - 4a(1-\bar{\xi})}}{2a}$$

$$\bar{u} = \frac{-1 + \sqrt{1 - a/2}}{a}$$

$$\bar{u} = -1/2 \quad \text{if} \quad a = 2$$

model closure coefficient

→ needed exact solution

⇒ no universal set of parameters for turbulence